



Aggarwal Kamikazes

Cutting & Coring Ltd.

GPR CONCRETE SCANNING · CORE CLEARANCE REPORT

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PROJECT
AKCC-2026-0431

OPERATOR
D. Cunningham

DATE
2026-05-29

PROJECT INFORMATION

Site	Cedar Crossing podium, 8800 Cambie, Vancouver BC	Equipment	GSSI StructureScan Mini XT (1.6 GHz)
Client	Northpoint Construction Ltd.	Structure	P1 / L2 / L3 suspended decks
Engineer of record	Pinnacle Structural Eng.	Coverage	9 grids · 22.4 m ² scanned
Scan date	2026-05-29	Project no.	AKCC-2026-0431

SCAN SUMMARY

GPR survey across three suspended decks to locate embedded reinforcement, post-tension tendons, electrical conduit and other services ahead of **nine proposed cores** (4"–8"). Reinforcement runs as a top mat at **~40-45 mm cover**; localized second-mat congestion was found on L2. Cores were assessed against a **50 mm minimum standoff** to any marked target. Results: **4 cleared, 3 caution, 2 no-go**.

CORE CLEARANCE FINDINGS

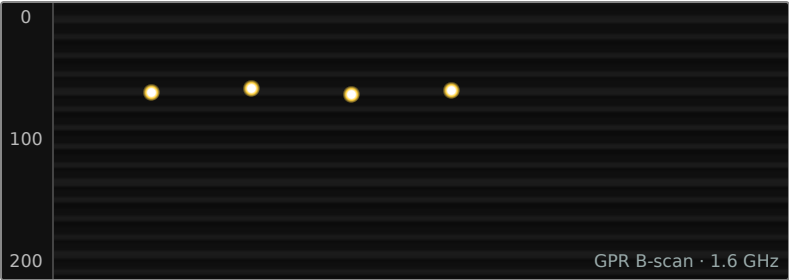
● 4 Safe ● 3 Caution ● 2 No-go

CORE	SIZE	LEVEL	VERDICT	CLEARANCE & RECOMMENDATION
A	4"	P1	SAFE	65 mm clear of all targets. Cleared to drill as marked.
B	6"	P1	CAUTION	PT tendon crosses within 30 mm — relocate 80 mm S and re-scan before drilling.
C	4"	P1	NO-GO	Directly over an energised conduit run. Do not drill; EOR redesign required.
D	8"	P1	SAFE	Large-diameter core clear at top mat; bottom mat at ~150 mm noted — drill depth limited to 130 mm.
E	4"	L2	CAUTION	Two-way mat congestion — best gap is 18 mm. Tight but workable; centre precisely on the mark.
F	5"	L2	SAFE	Hydronic radiant tubing mapped and avoided; 55 mm clear. Cleared as marked.
G	4"	L2	NO-GO	Unidentified metallic object at ~90 mm directly below. No-go pending intrusive verification.
H	6"	L2	CAUTION	Within 300 mm of slab edge; possible void beneath. Reduced confidence — proceed slowly, verify on site.
J	4"	L3	SAFE	Single top mat, generous spacing. 70 mm clear in all directions. Cleared as marked.

SCAN LOCATIONS — A & B

Grid A · P1 Bay 1 mid-span

0.9 m × 0.9 m · depth 0–200 mm · Core A



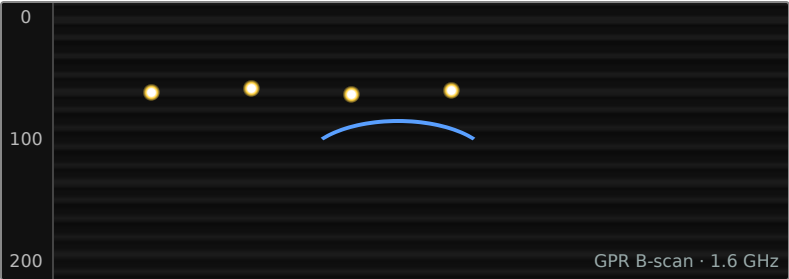
■ Rebar — top mat, ~45 mm

SAFE

Observation. Clear cone at the proposed centre — no targets within 65 mm in any direction. Uniform top mat at ~45 mm cover. Safe to drill as marked.

Grid B · P1 Bay 2 column line

0.9 m × 0.9 m · depth 0–200 mm · Core B



■ Rebar — top mat

■ Post-tension tendon

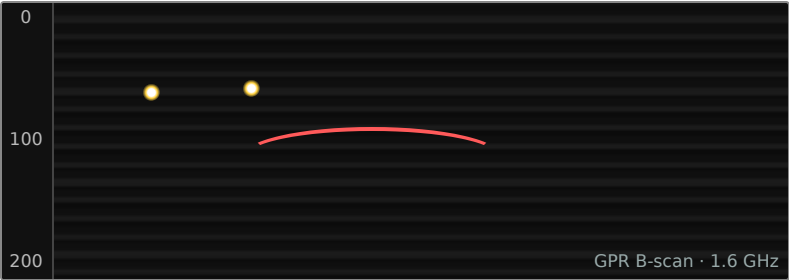
CAUTION

Observation. A post-tension tendon crosses on a diagonal within 30 mm of the proposed centre. Relocate the core 80 mm south, clear of the tendon envelope, and re-scan before drilling.

SCAN LOCATIONS — C & E

Grid C · P1 Bay 2 wall line

0.9 m × 0.9 m · depth 0–200 mm · Core C



0
100
200

Rebar — top mat
Energised conduit / no-scan zone

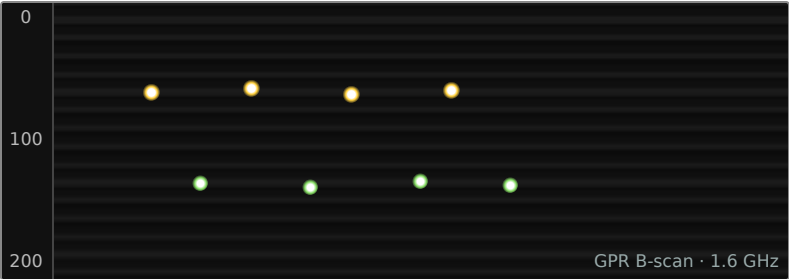
NO-GO

GPR B-scan · 1.6 GHz

Observation. Energised electrical conduit detected directly beneath the proposed centre at ~110 mm. No-go — do not drill; refer to the EOR for a redesigned core position.

Grid E · L2 Bay 4 mid-span

0.9 m × 0.9 m · depth 0–200 mm · Core E



0
100
200

Rebar — top mat, ~40 mm
Rebar — second mat, ~95 mm

CAUTION

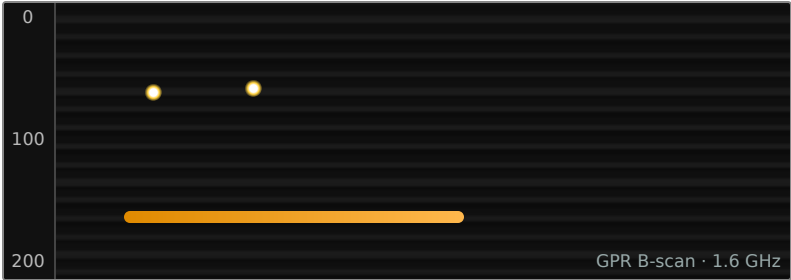
GPR B-scan · 1.6 GHz

Observation. Heavy two-way reinforcement — top and second mats both present. The best clear gap at the mark is 18 mm. Workable for a 4" core if centred precisely; flag the tight tolerance to the operator.

SCAN LOCATIONS — F & G

Grid F · L2 mechanical room

0.9 m × 0.9 m · depth 0–200 mm · Core F



0
100
200

Rebar — top mat
Hydronic radiant tubing

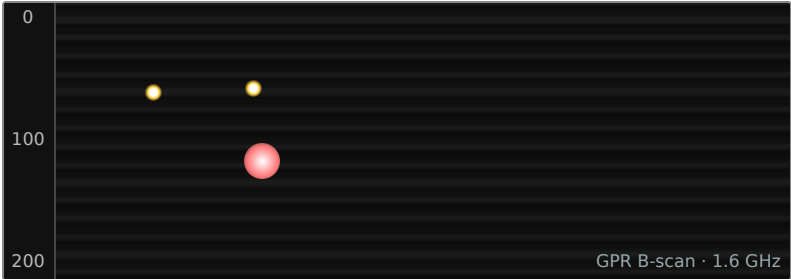
SAFE

GPR B-scan · 1.6 GHz

Observation. Radiant floor-heating tubing mapped along the south of the grid and avoided. Proposed centre is 55 mm clear of both rebar and tubing. Cleared to drill as marked.

Grid G · L2 Bay 5 wall line

0.9 m × 0.9 m · depth 0–200 mm · Core G



0
100
200

Rebar — top mat
Unidentified metallic object

NO-GO

GPR B-scan · 1.6 GHz

Observation. A strong, isolated metallic reflection sits at ~90 mm directly below the proposed centre — not consistent with rebar spacing. No-go until its nature is confirmed by intrusive verification (e.g. small inspection hole).

SCAN LOCATIONS — H

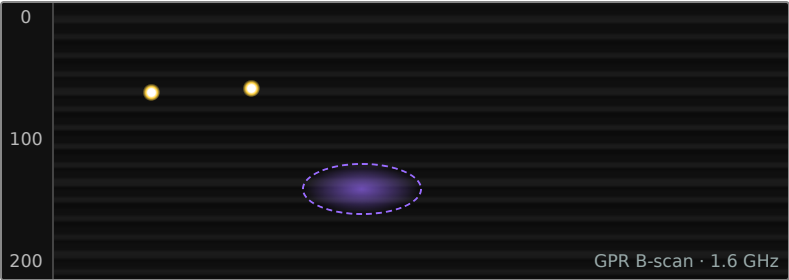
Grid H · L2 slab edge

0.9 m × 0.9 m · depth 0–200 mm · Core H

0

100

200

A GPR B-scan image showing a cross-section of a concrete slab. The vertical axis on the left represents depth in millimeters, with markers at 0, 100, and 200. The horizontal axis represents distance. Two bright, yellowish-white spots near the surface (around 20 mm depth) represent the top mat of rebar. A horizontal, purple, oval-shaped region with a dashed outline is located at a depth of approximately 120-150 mm, indicating a suspected void or delamination. The text 'GPR B-scan · 1.6 GHz' is visible in the bottom right corner of the image area.

■ Rebar — top mat

■ Suspected void / delamination

CAUTION

Observation. Core sits ~300 mm from a slab edge with a low-amplitude zone suggesting a void or delamination beneath. Confidence reduced. Proceed slowly with a pilot; stop and reassess if the bit breaks through.

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METHODOLOGY & STANDARD NOTES

1. Survey performed with a GSSI StructureScan Mini XT (1.6 GHz) on orthogonal 50 mm line spacing.
2. Depths are estimates based on an assumed dielectric for cured concrete and are $\pm 15\%$.
3. Targets marked on the slab in matching colours supersede this report where any discrepancy exists.
4. Clearances reference the marked grid origin; verify on site before cutting.
5. "No-go" locations require resolution with the engineer of record before any drilling.

LIMITATIONS

Limitations. GPR results are interpretive and depend on moisture, cover and congestion. Non-metallic and deeply embedded targets, and objects shadowed by dense upper steel, may not be detected. Aggarwal Kamikazes Cutting & Coring Ltd. accepts no liability for drilling outside the cleared zones described above. This report is valid for the slabs, locations and date stated.

SCANNED BY

D. Cunningham · GPR Technician

DATE

2026-05-29

REVIEWED BY

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